

Bond disorder in non-collinear antiferromagnets

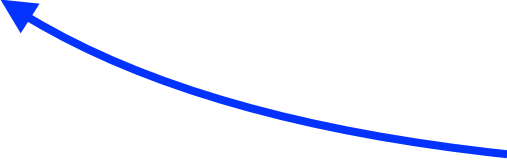
Santanu Dey and Matthias Vojta
TU Dresden



Bond-disordered Heisenberg magnets

$$H = \sum_{i,j} J_{ij} \mathbf{S}_i \cdot \mathbf{S}_j$$

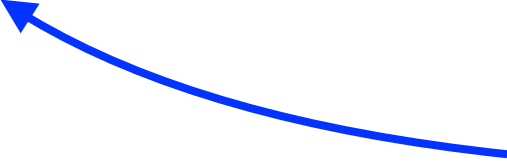
Disordered couplings



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Disordered couplings



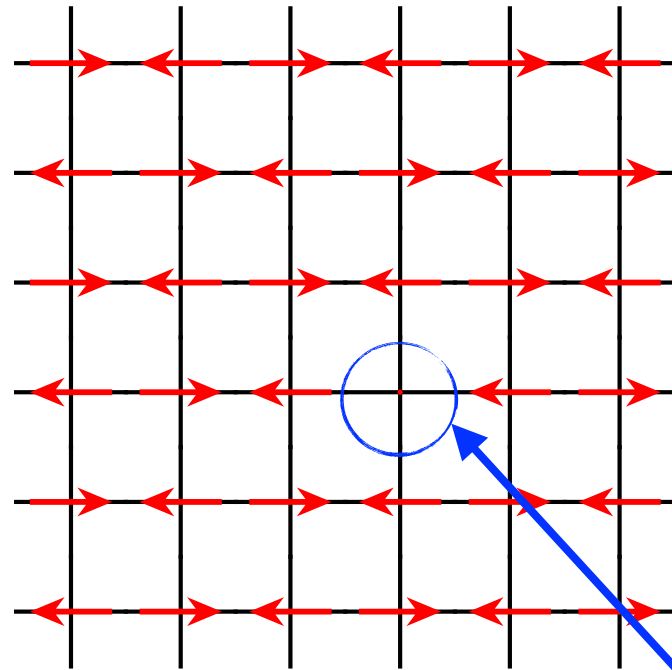
- Except from lattice translational invariance, they do not break any other symmetries of the Hamiltonian.

Bond disorder effects can be crucial for non-collinear antiferromagnets.

Heisenberg antiferromagnet

Vacancy

Collinear square lattice AFM



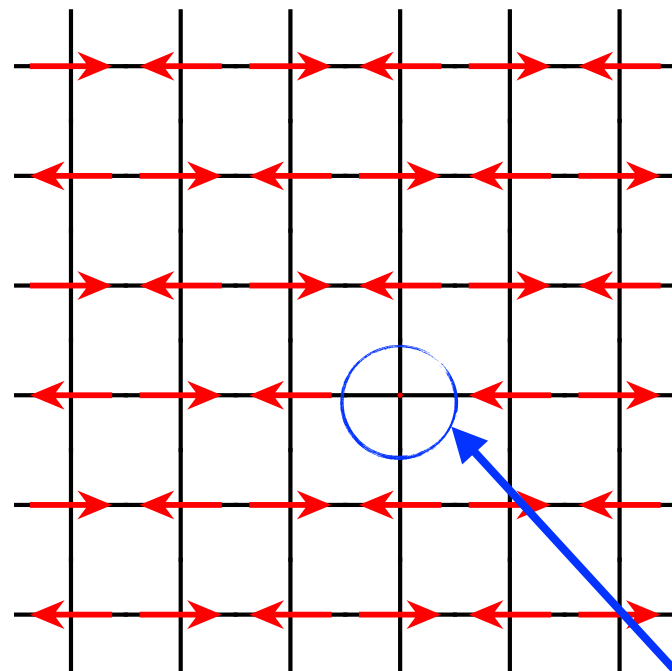
Néel order

Absent spin

Heisenberg antiferromagnet

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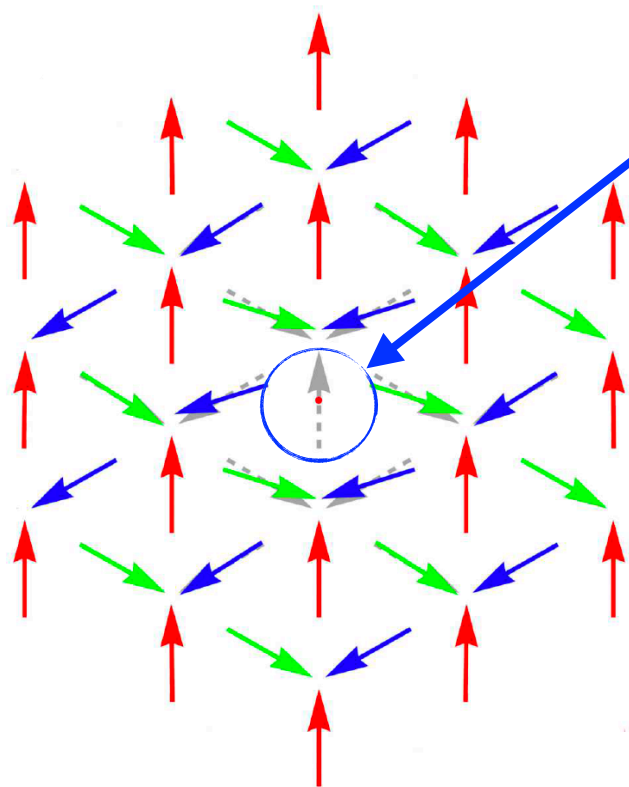


Néel order



Absent spin

Non-collinear triangular lattice AFM



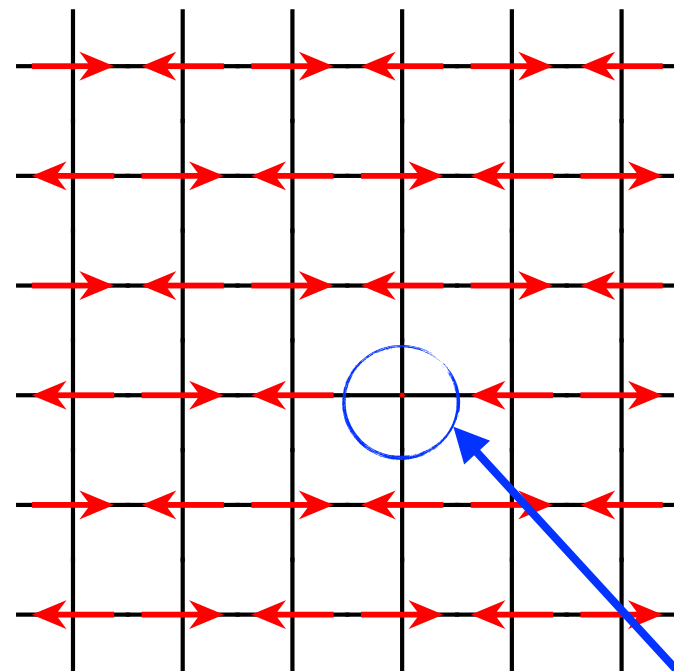
Vacancy induced distortion



Heisenberg antiferromagnet

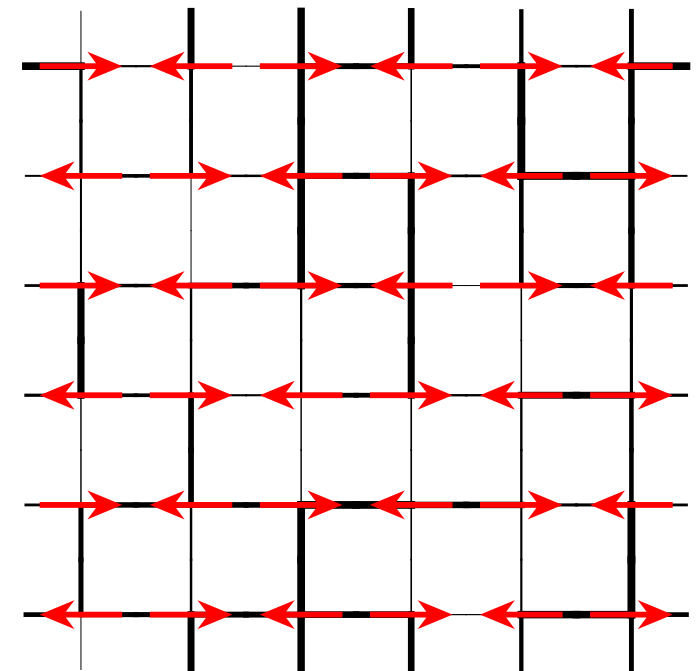
Collinear square lattice AFM

Vacancy



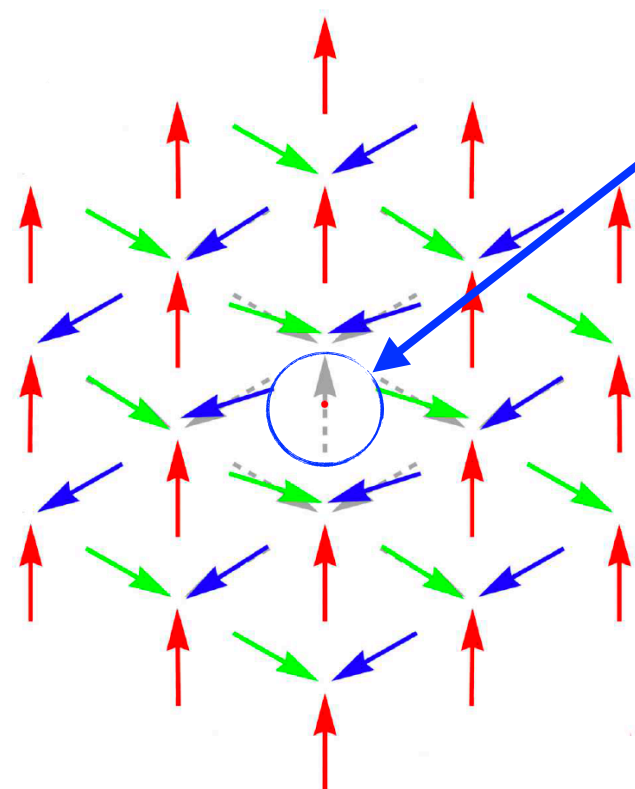
Néel order

Bond disorder



Non-collinear triangular lattice AFM

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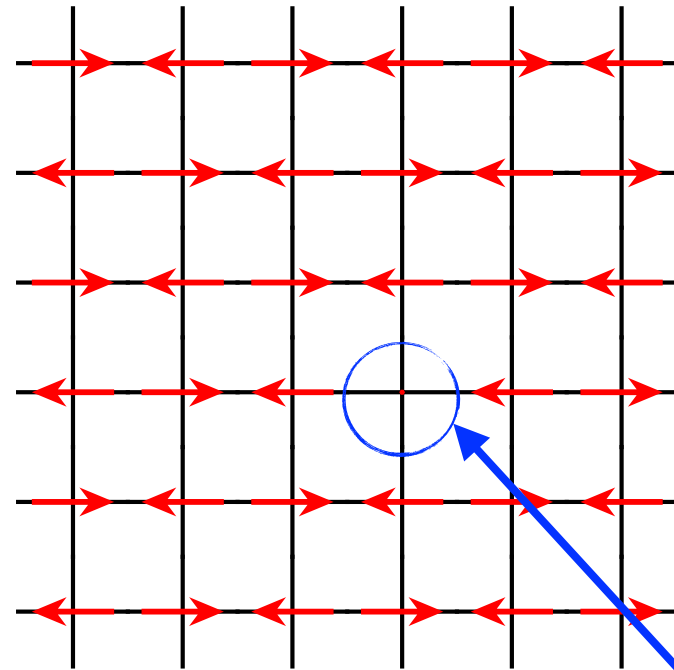


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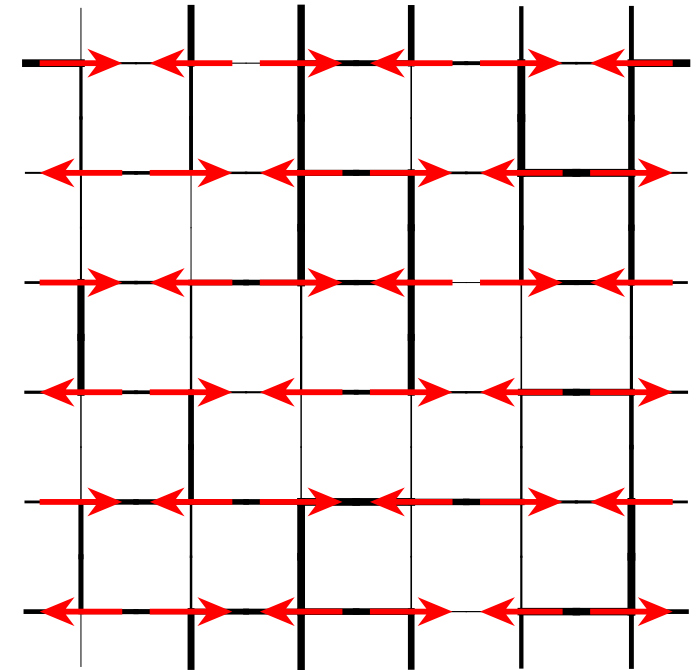
Collinear square-lattice AFM

Vacancy



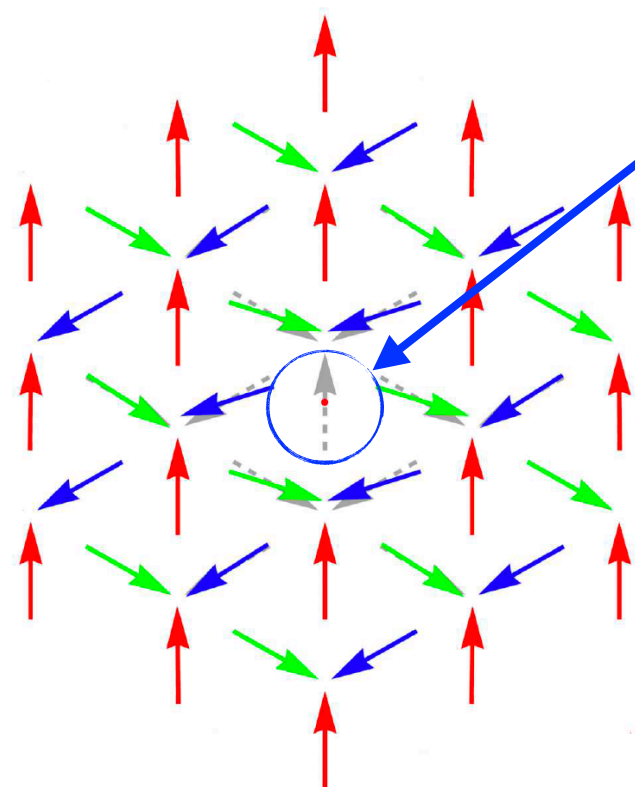
Néel order

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Non-collinear triangular lattice AFM

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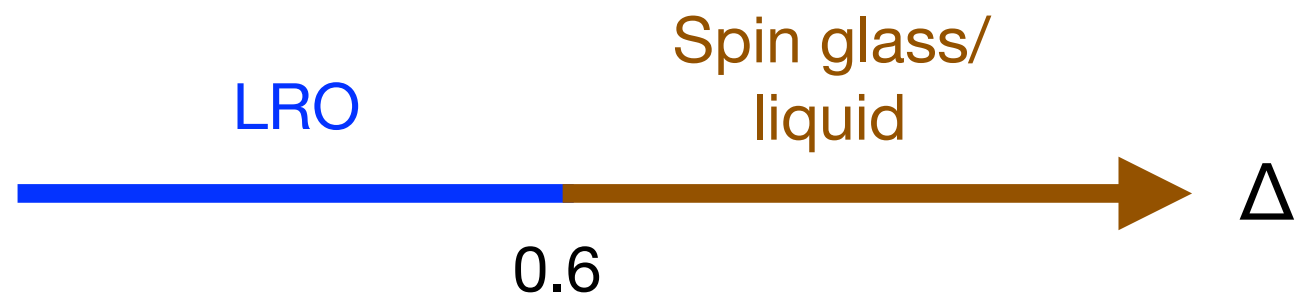
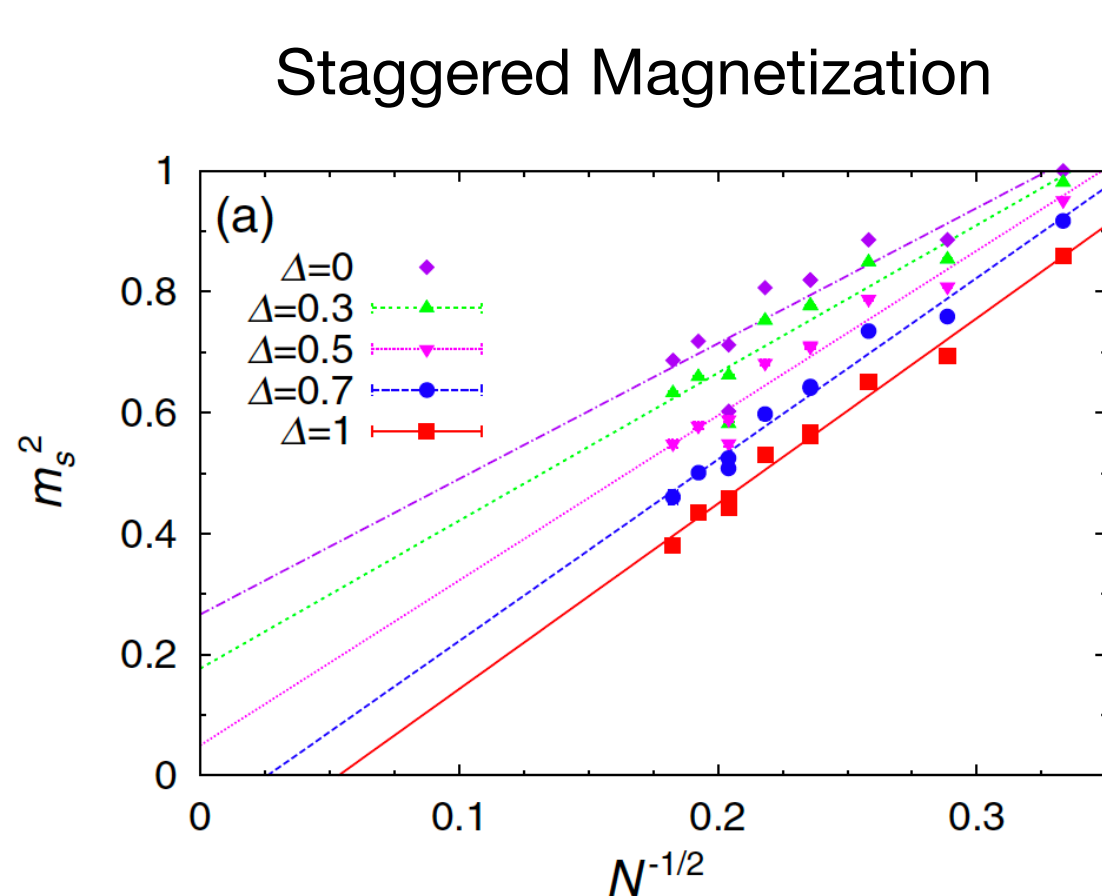
Is the long range order (LRO) affected by bond disorder?

Numerical simulation

- ED for a disordered triangular lattice Heisenberg model,

$$H = J \sum_{\langle i,j \rangle} (1 + \Delta \alpha_{ij}) \mathbf{S}_i \cdot \mathbf{S}_j$$

with α_{ij} drawn from a random uniform distribution, $\alpha_{ij} \in [-1, 1]$



Extrapolated finite size data signals the destruction of LRO at a finite value of disorder magnitude.

Isolated bond defect

- Classical triangular lattice Heisenberg antiferromagnet with bond defect at the center.

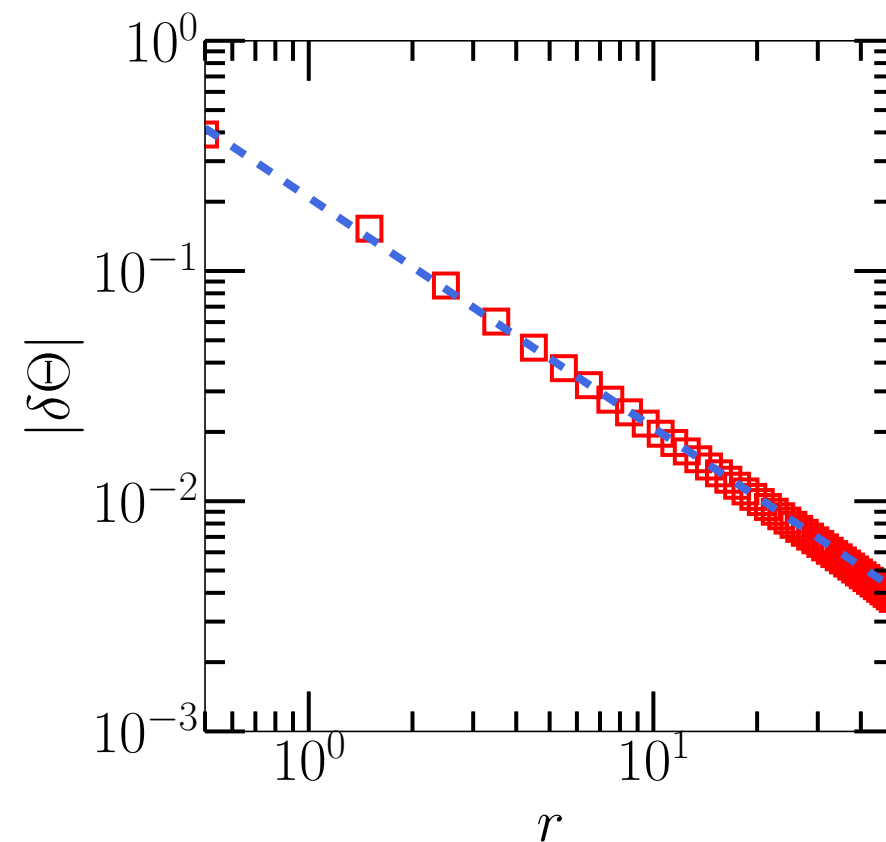
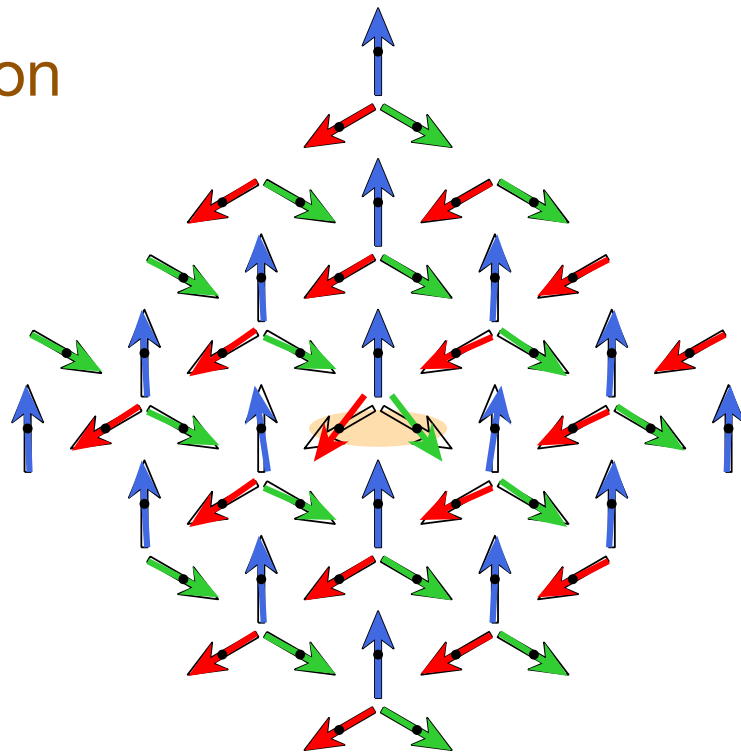
$$H = \sum_{\langle i,j \rangle} J_{ij} \mathbf{S}_i \cdot \mathbf{S}_j \quad \text{with} \quad J_{ij} = \begin{cases} J - \delta J. & \text{if } (i, j) = (1, 0) \\ J, & \text{otherwise} \end{cases}$$

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Bond direction
 $\delta = (1, 0)$



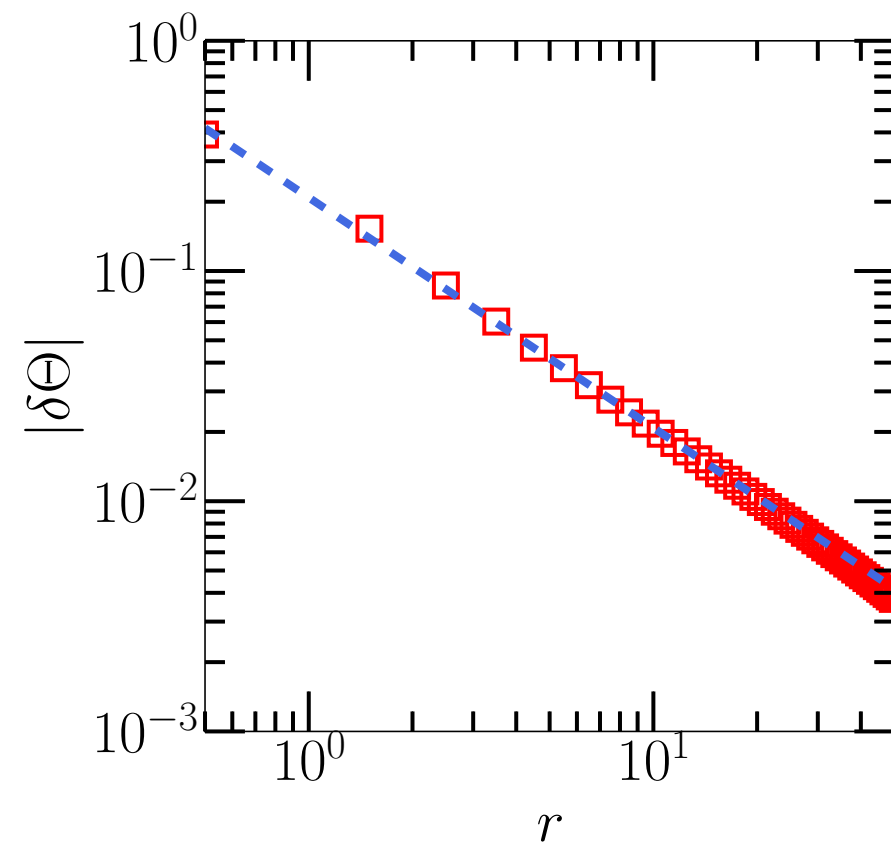
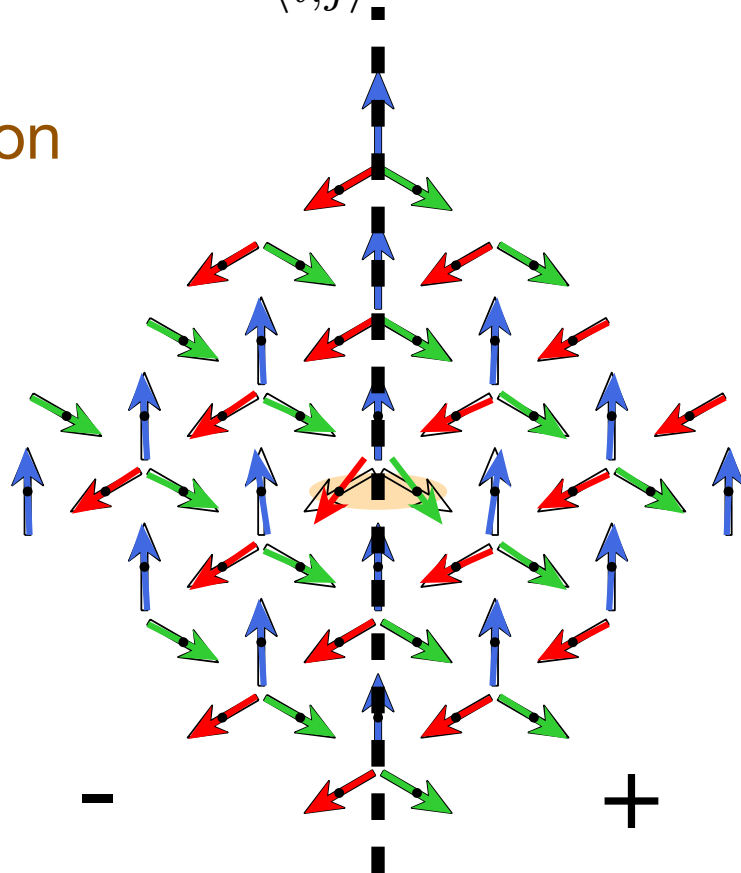
Spins away from the defect are rotated from their 120-degree orientation

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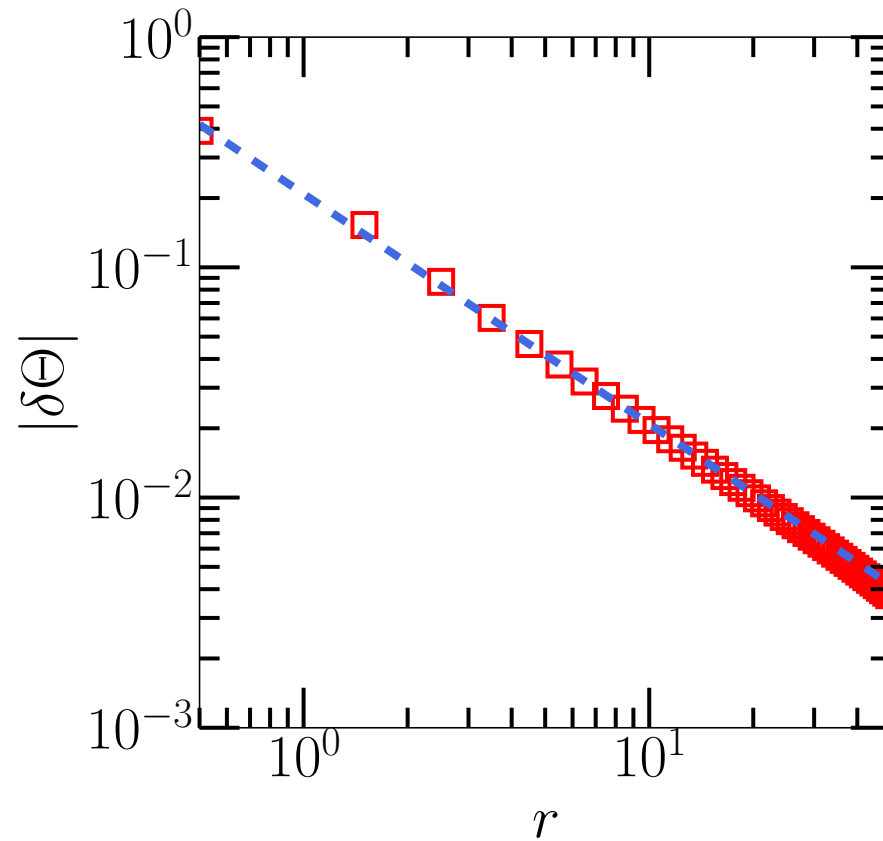
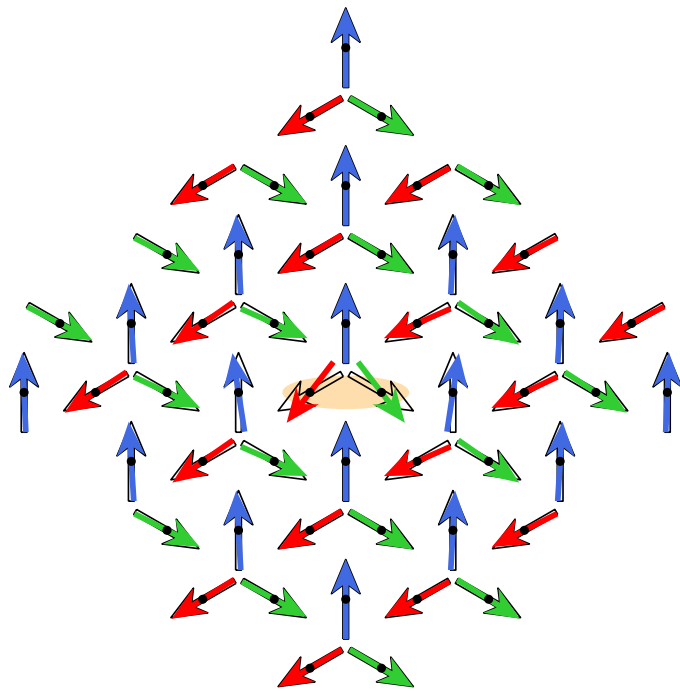


Spins away from the defect are rotated from their 120-degree orientation

2D dipolar profile $\delta\Theta(r) \propto \frac{\cos \theta}{r}$ ← Angular dependence dictated by dipole moment δ

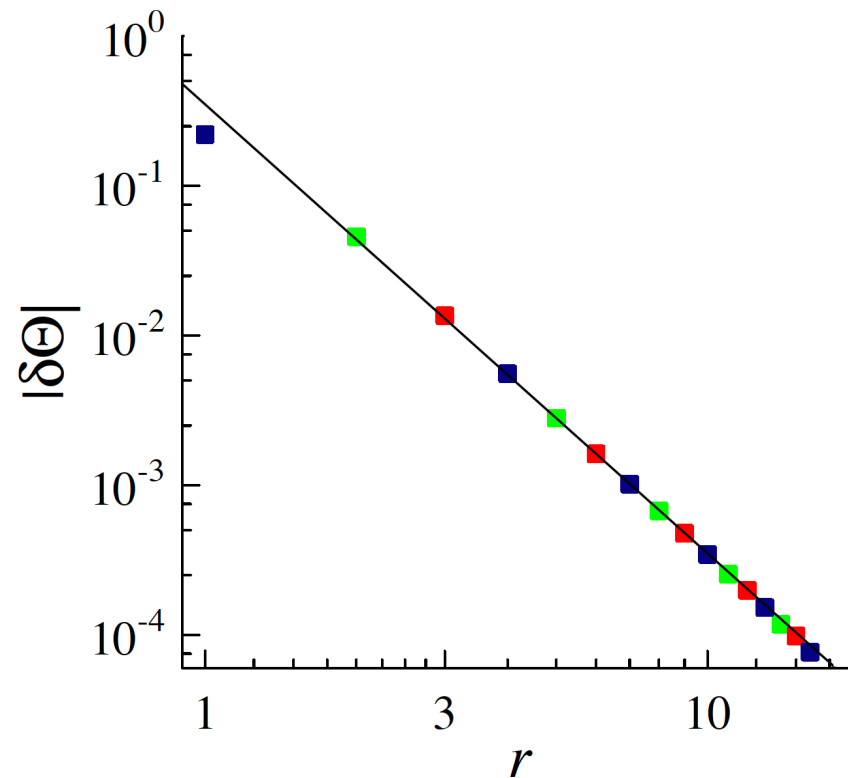
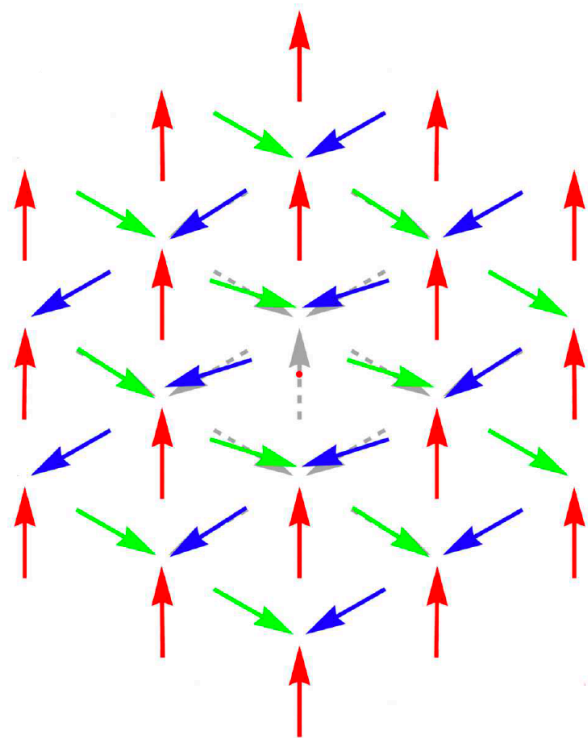
Bond defect vs vacancy

Bond defect



$$\delta\Theta(r) \propto \frac{1}{r}$$

Vacancy



$$\delta\Theta(r) \propto \frac{1}{r^3}$$

A. Wollny, Lars Fritz, and M. Vojta,
Phys. Rev. Lett. 107, 137204
(2011)

Single-defect response

- Bond-defect $\delta J(\mathbf{r}')$ along the bond direction $\delta(\mathbf{r}')$ the angle correction $\delta\Theta(\mathbf{r}) \sim$ transverse local spin response.

$$\langle S_r^\perp \rangle \propto \frac{\delta J(\mathbf{r}')}{J} \sum_{\mathbf{q}} \mathbf{q} \cdot \delta(\mathbf{r}') \frac{e^{i\mathbf{q} \cdot (\mathbf{r} - \mathbf{r}')}}{q^2}$$

Dipole structure factor
due to the bond defect

Transverse spin correlation

$$\langle\langle \mathbf{S}_{\mathbf{q}}^\perp; \mathbf{S}_{-\mathbf{q}}^\perp \rangle\rangle$$

Dipolar response in momentum space

Concentration of defects

- Small disorder → Superposition of bond defects

$$\langle S_r^\perp \rangle = \sum_{\mathbf{r}'} \sum_{\mathbf{q}} \frac{\delta J(\mathbf{r}')}{J} \frac{\mathbf{q} \cdot \boldsymbol{\delta}(\mathbf{r}')}{q^2} e^{i\mathbf{q} \cdot (\mathbf{r} - \mathbf{r}')}$$

Sum over bond defects



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Sum over bond defects

- For random distribution of bonds:

$$\overline{\delta J(\mathbf{r})} = 0$$

$$\overline{\delta J(\mathbf{r}) \delta J(\mathbf{r}')} = \delta J^2 \delta^{(d)}(\mathbf{r} - \mathbf{r}')$$

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Local
fluctuation
amplitude

$$\langle \overline{(S^\perp)^2} \rangle = \left(\frac{\delta J}{J} \right)^2 \int \frac{d^d q}{(2\pi)^d} \frac{1}{q}$$

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→ Divergent in $d = 2$

Breakdown of LRO

$$\langle \overline{(S^\perp)^2} \rangle = \left(\frac{\delta J}{J} \right)^2 \int \frac{d^d q}{(2\pi)^d} \frac{1}{q}$$



Infrared divergence

Infrared divergence of fluctuation
in $d = 2$ implies destruction of
LRO

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Infrared divergence

$1/\xi$

Infrared divergence of fluctuation
in $d = 2$ implies destruction of
LRO

For $\delta J \rightarrow 0$ this implies an exponential divergence of
the correlation length.

$$\xi \propto e^{\frac{1}{(\delta J/J)^2}}$$

Contrast to dilution

Bond defects

Single defect $\rightarrow$ slow decay

$$\langle S^\perp \rangle \propto \frac{1}{r}$$

in $d = 2$



LRO is destroyed for small bond disorder

$$\overline{\langle (S^\perp)^2 \rangle} \rightarrow \infty$$

Vacancies

Single vacancy $\rightarrow$ fast decay

$$\langle S^\perp \rangle \propto \frac{1}{r^3}$$

in $d = 2$



LRO is stable against small concentration of vacancies

Competing effects

S. Chakravarty, B. I. Halperin, and D. R. Nelson, Phys. Rev. B 39, 2344 (1989)

- Finite T - 2D:

Thermal correlation length also diverges exponentially $\xi \propto e^{\frac{1}{T/J}}$

Crossover
temperature

$$\left(\frac{T^*}{J}\right) \simeq \left(\frac{\delta J}{J}\right)^2$$

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- Layered magnet:

Finite interlayer coupling $J^\perp$ stabilizes LRO.

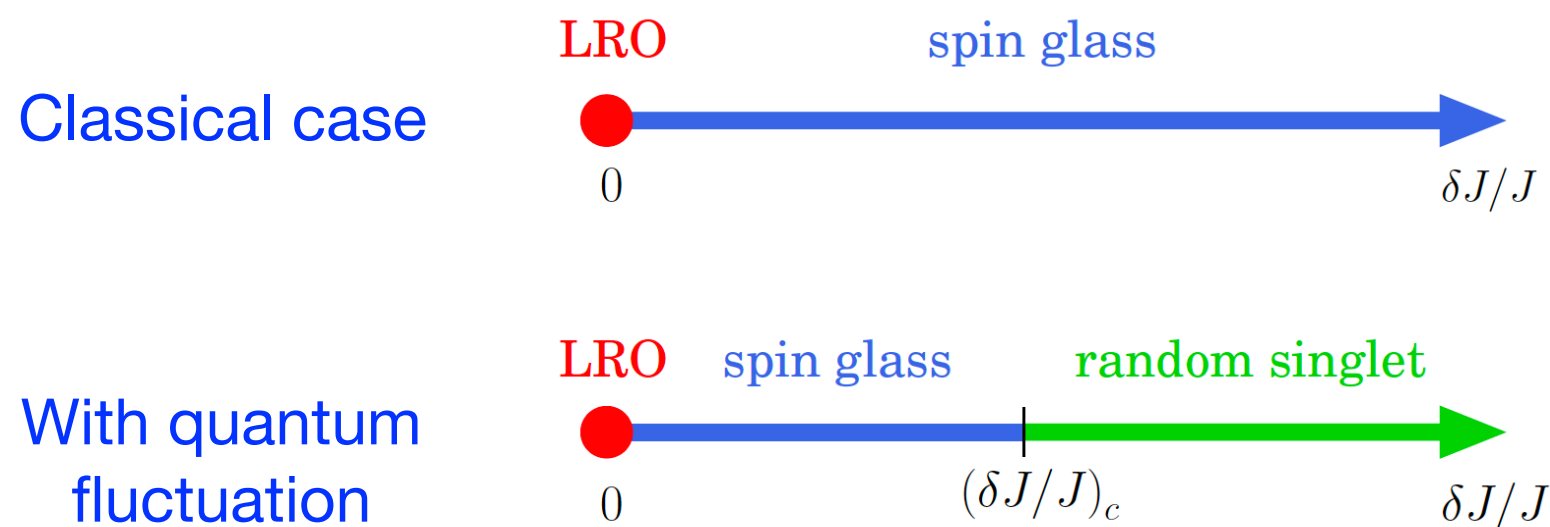
$$\left(\frac{J^\perp}{J}\right) > \left(\frac{\delta J}{J}\right)^2$$

Stability criterion

Summary

- Bond defects in non-collinear magnets produce dipolar distortion with slow long distance decay.
 $\sim \frac{1}{r}$ in $d = 2$
- In $d = 2$ or less, concentration of such defects destroy LRO even for very small disorder magnitude.
- The resulting ground state has short ranged correlation which exponentially diverge with disorder magnitude.

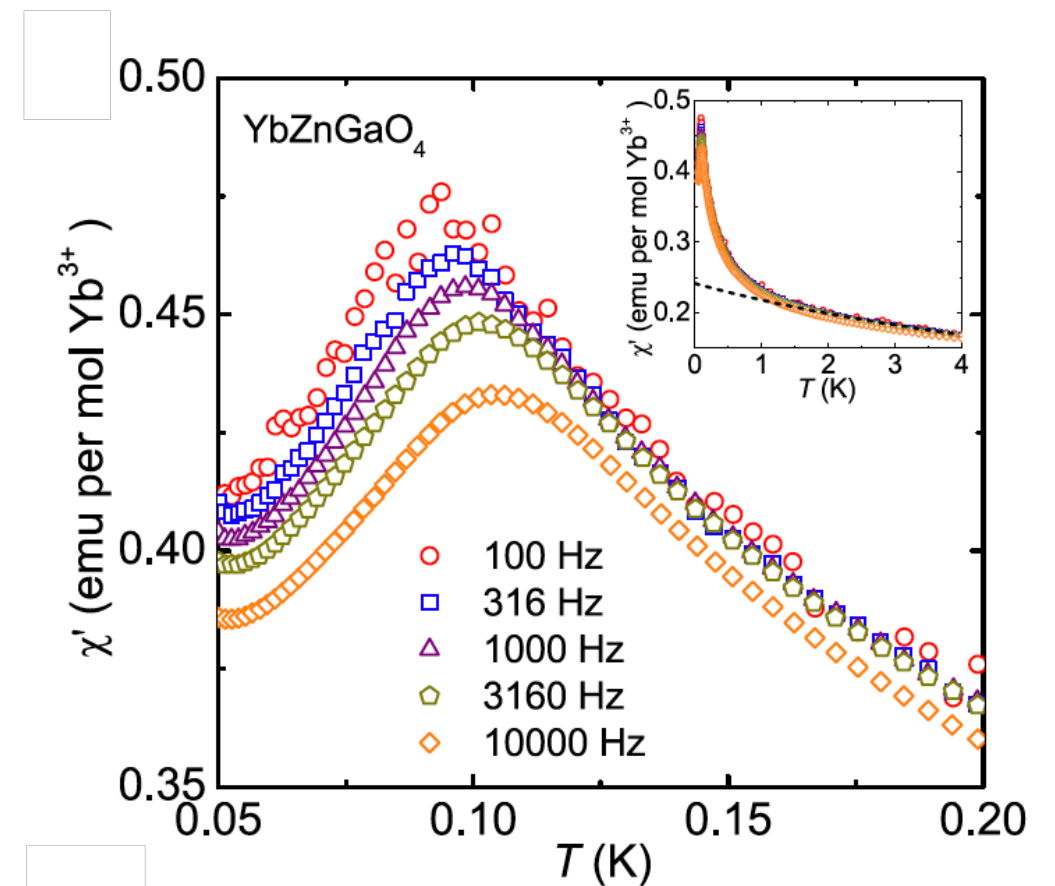
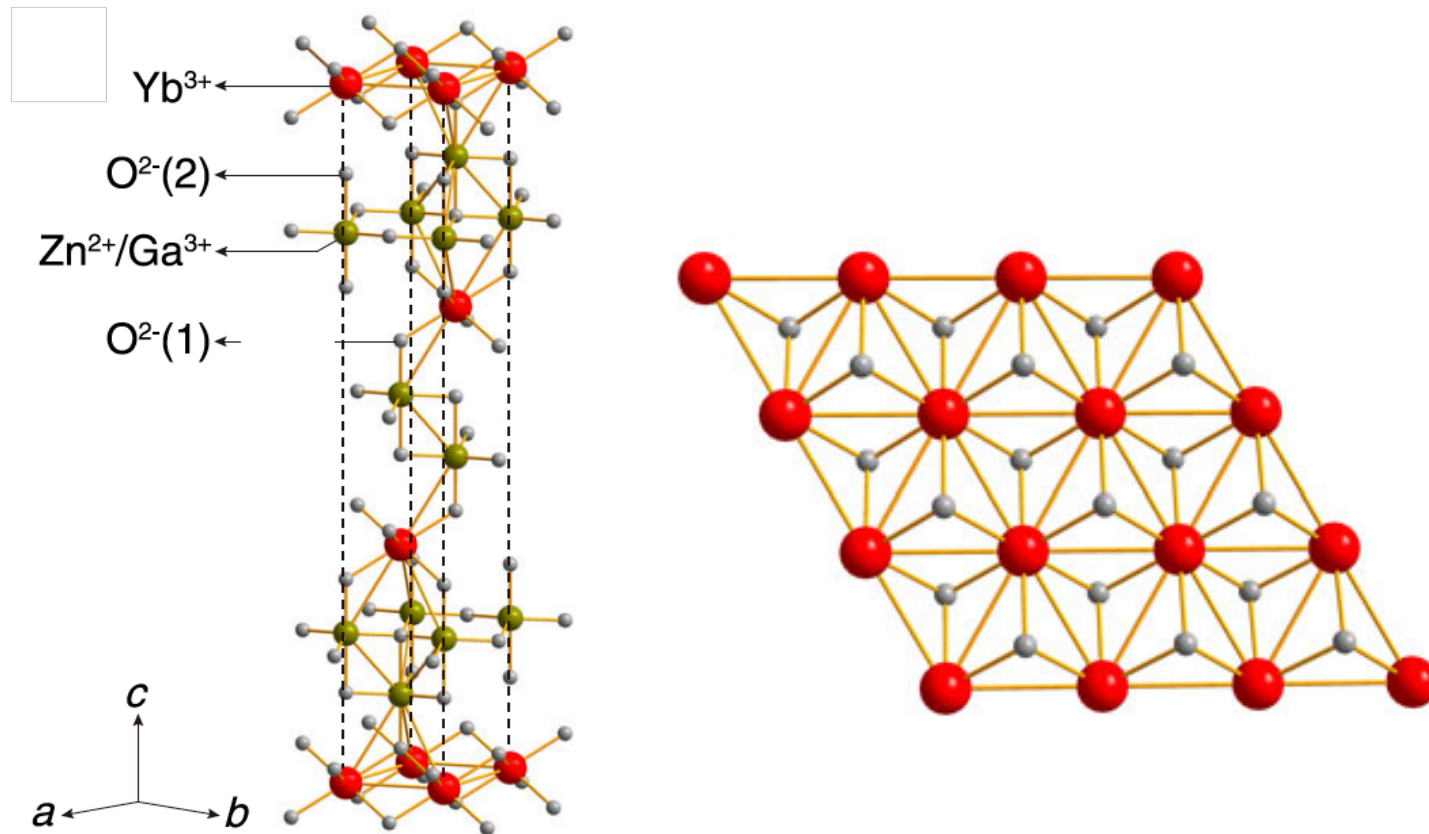
$$\xi \propto e^{\left(\frac{J}{\delta J}\right)^2}$$



Extra slides

Experiment

- Experiment on 2D triangular lattice compound YbZnGaO_4 revealed a spin-glass ground state



Low-frequency response indicating glassy behavior:

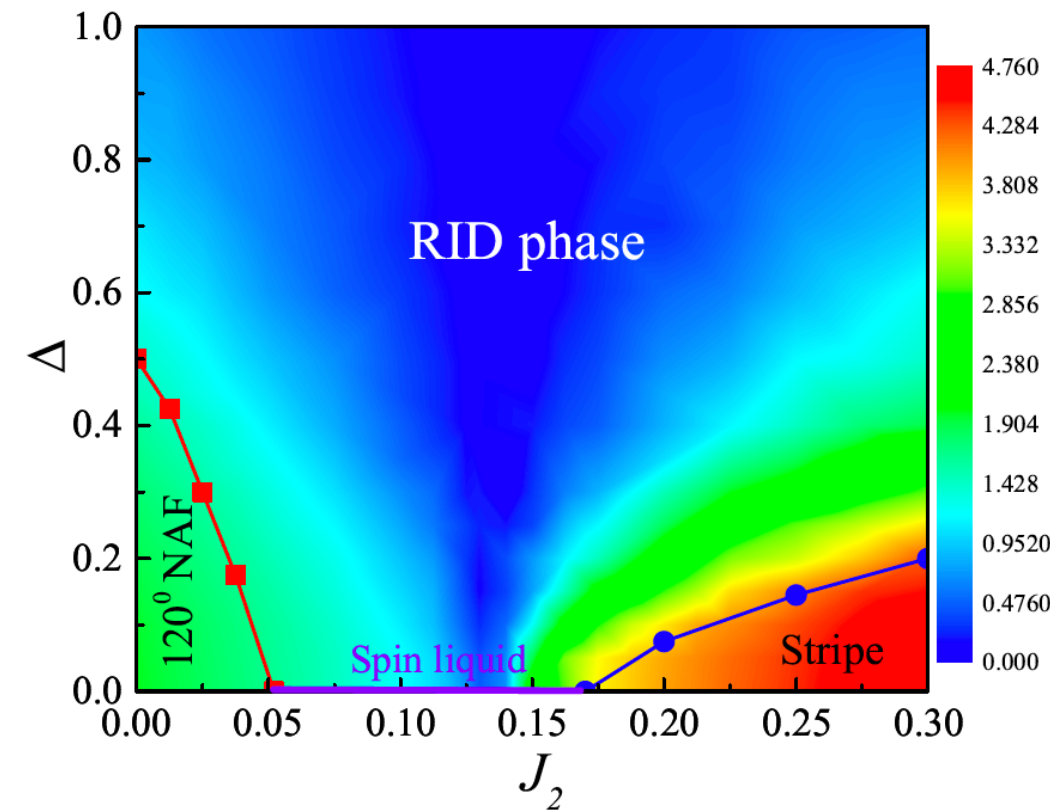
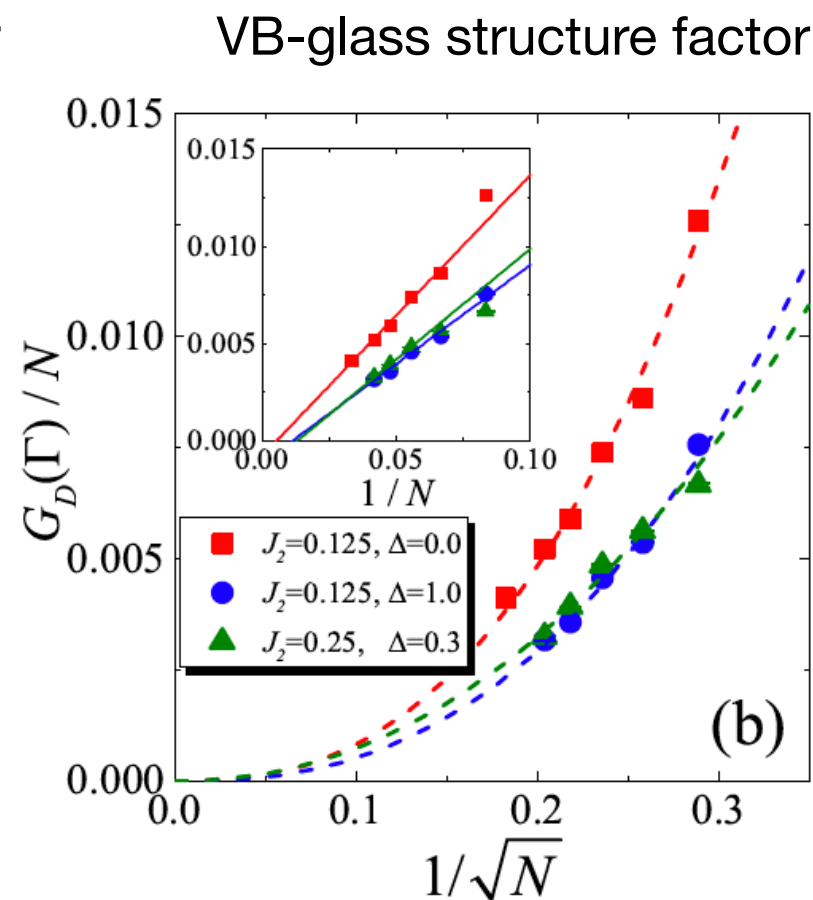
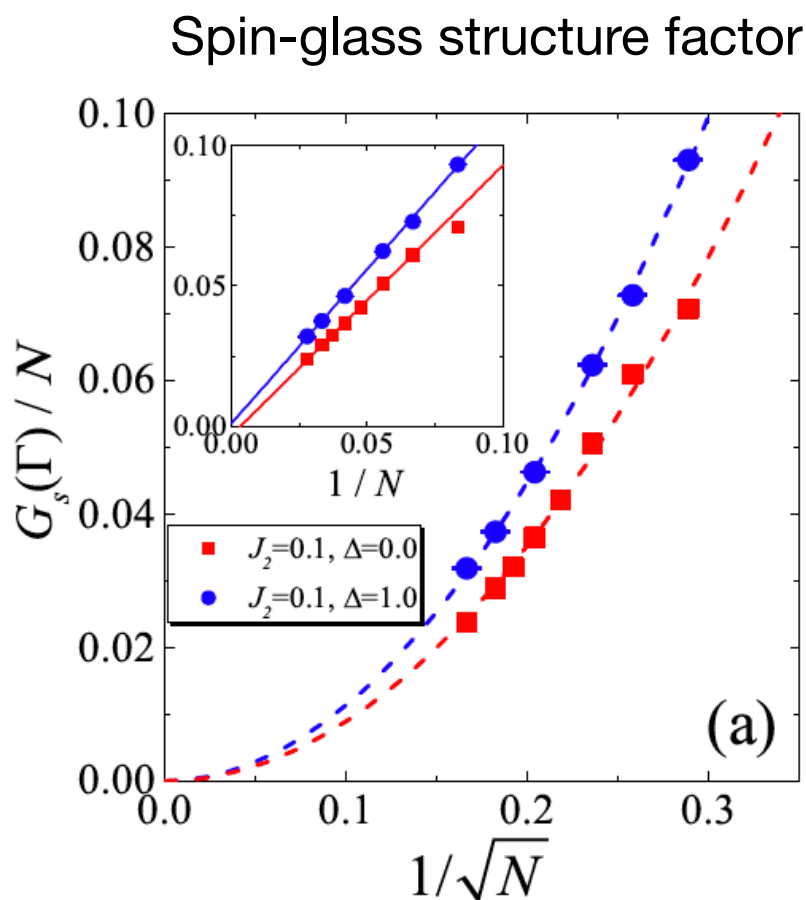
[Z. Ma et. al, Phys. Rev. Lett. 120, 087201 \(2018\)](#)

Numerical simulation

- ED and DMRG analysis of the disordered J_1 - J_2 triangular lattice Heisenberg model,

$$H = J_1 \sum_{\langle i,j \rangle} (1 + \Delta \alpha_{ij}) \mathbf{S}_i \cdot \mathbf{S}_j + J_2 \sum_{\langle\langle i,j \rangle\rangle} (1 + \Delta \beta_{ij}) \mathbf{S}_i \cdot \mathbf{S}_j$$

with $\alpha_{ij}, \beta_{ij} \in [-1, 1]$

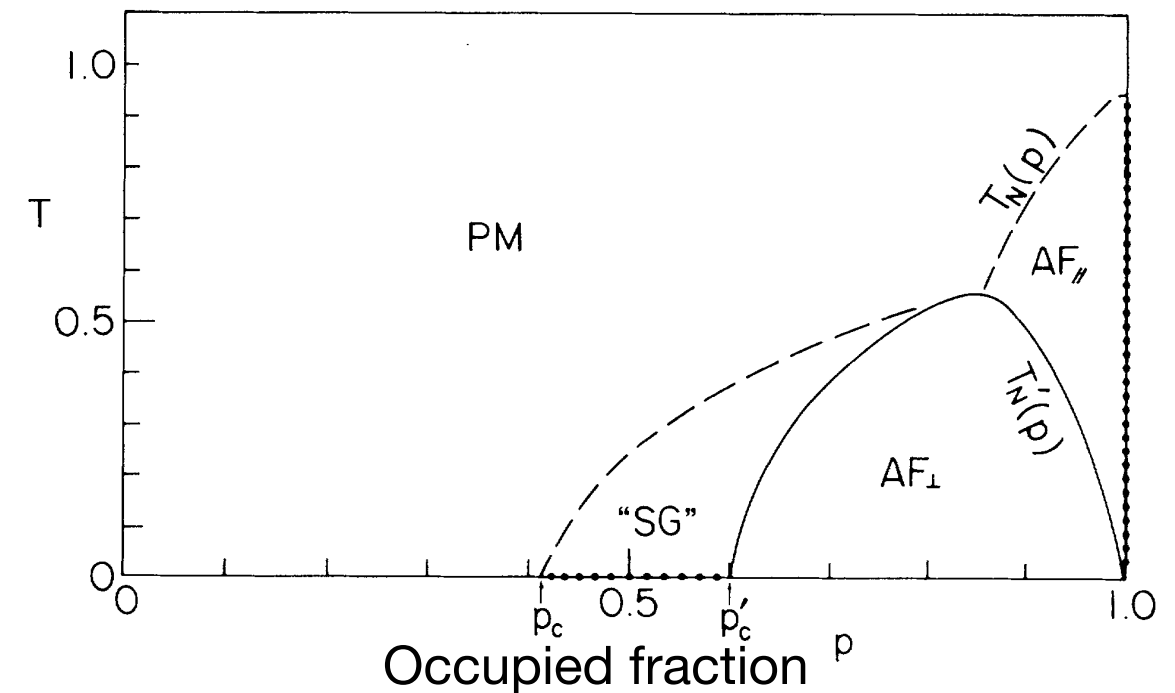


Indication of **Randomness Induced Disordered** phase without magnetic, spin-glass or valence-bond glass ordering

H. Wu et. al, [arXiv:1802.00935](https://arxiv.org/abs/1802.00935) (2018)

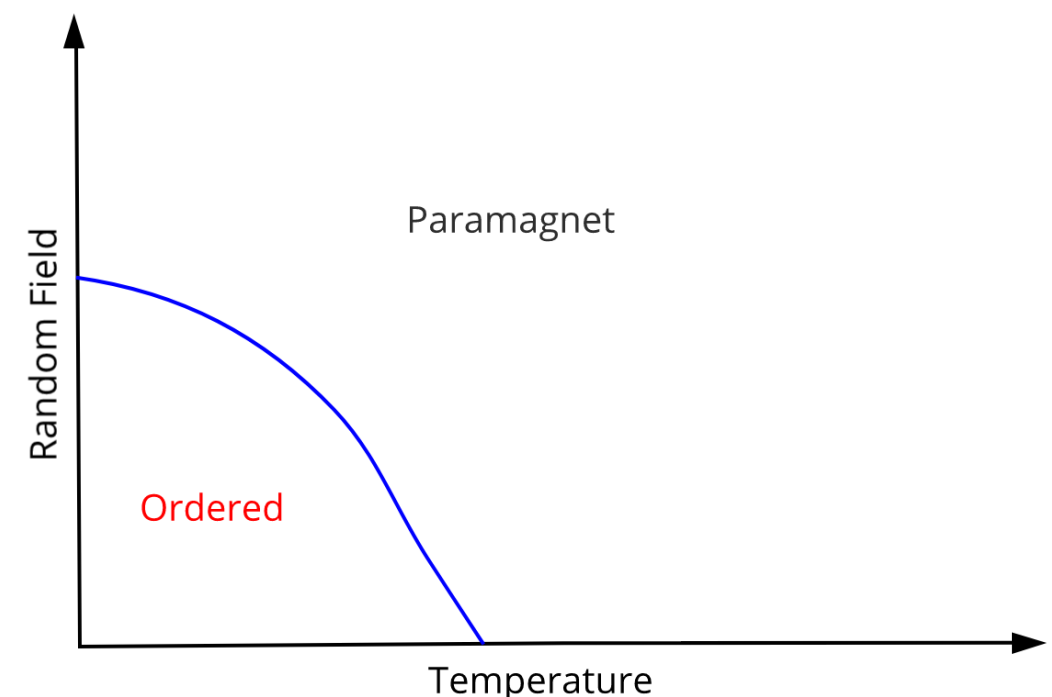
Quenched disorder affecting long range order

- Random site-dilution can induce glassy behavior or long range magnetic order depending on competing interactions of a model.



The phase diagram of site diluted J_1 - J_2 XY antiferromagnet:
[C. L. Henley, Phys. Rev. Lett. 62, 2056 \(1989\)](#)

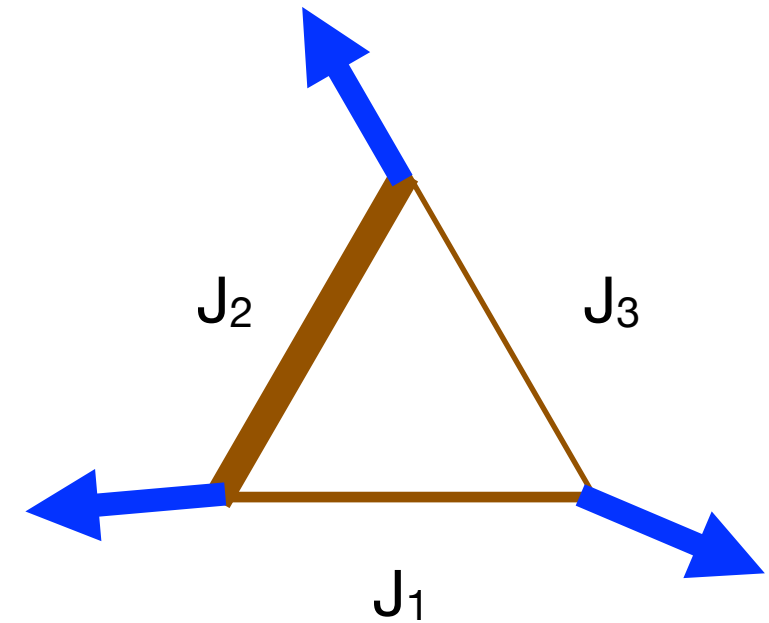
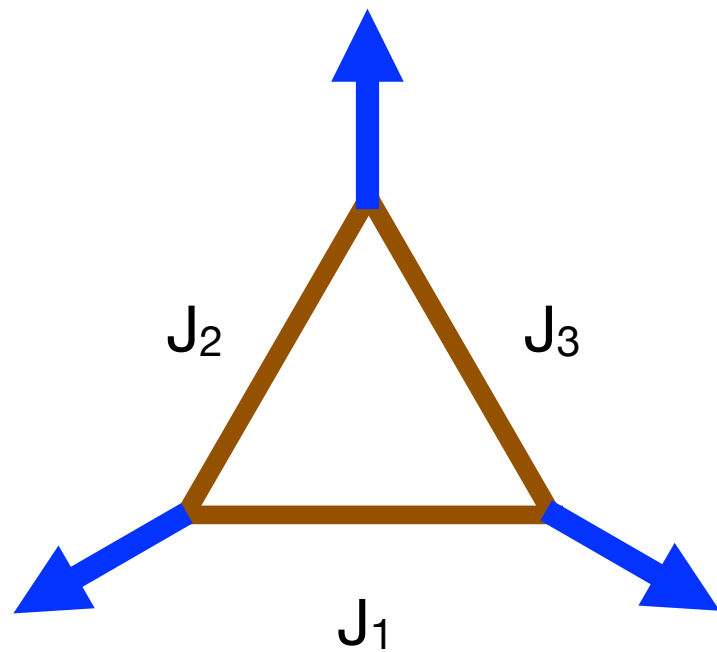
- Infinitesimal random fields destroy long range order (LRO) below or at the lower critical dimension
- Imry-Ma argument:** $d_c = 2$ for discrete symmetries and $d_c = 4$ for continuous symmetries



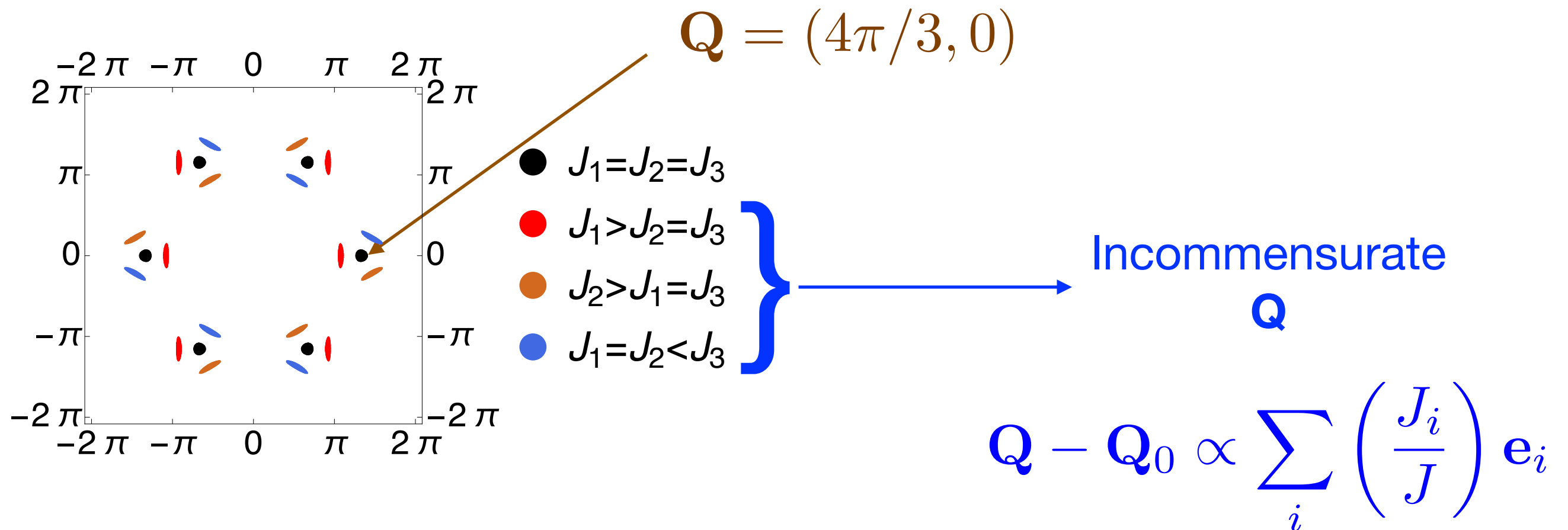
The theoretical phase diagram of 3D random-field Ising antiferromagnet

Effect on the state

Bond disorder turns the triangles locally anisotropic



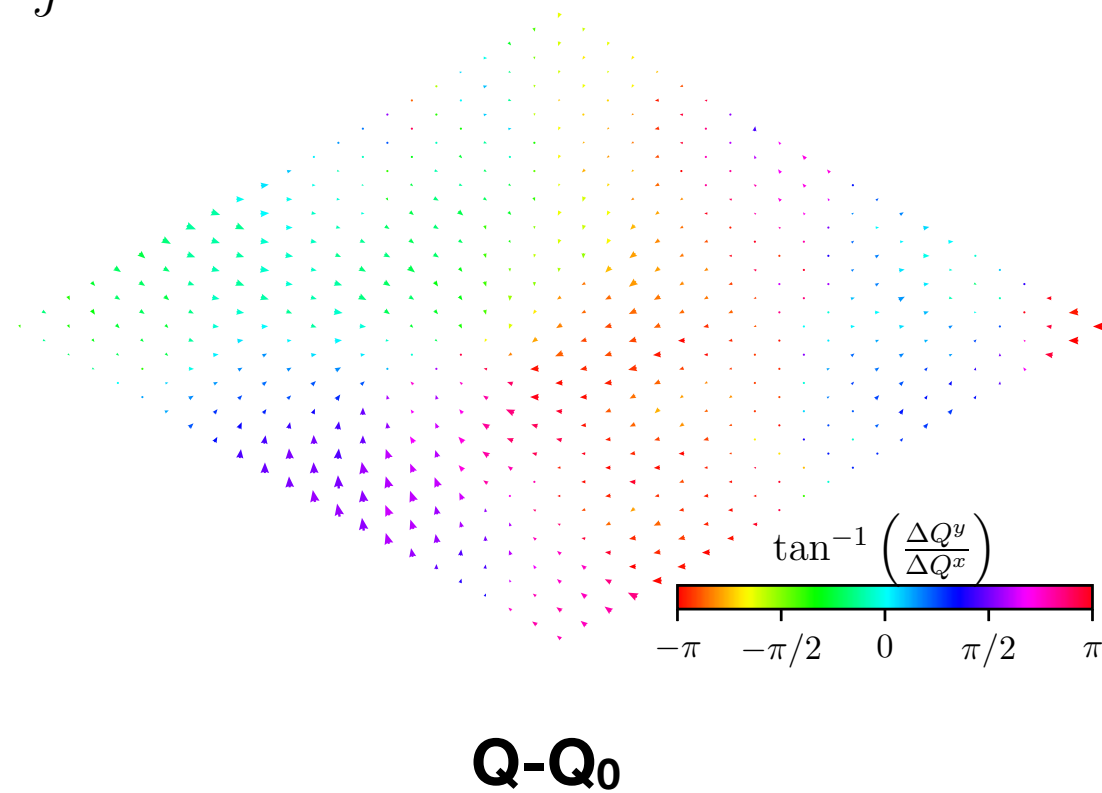
Globally anisotropic triangular lattice



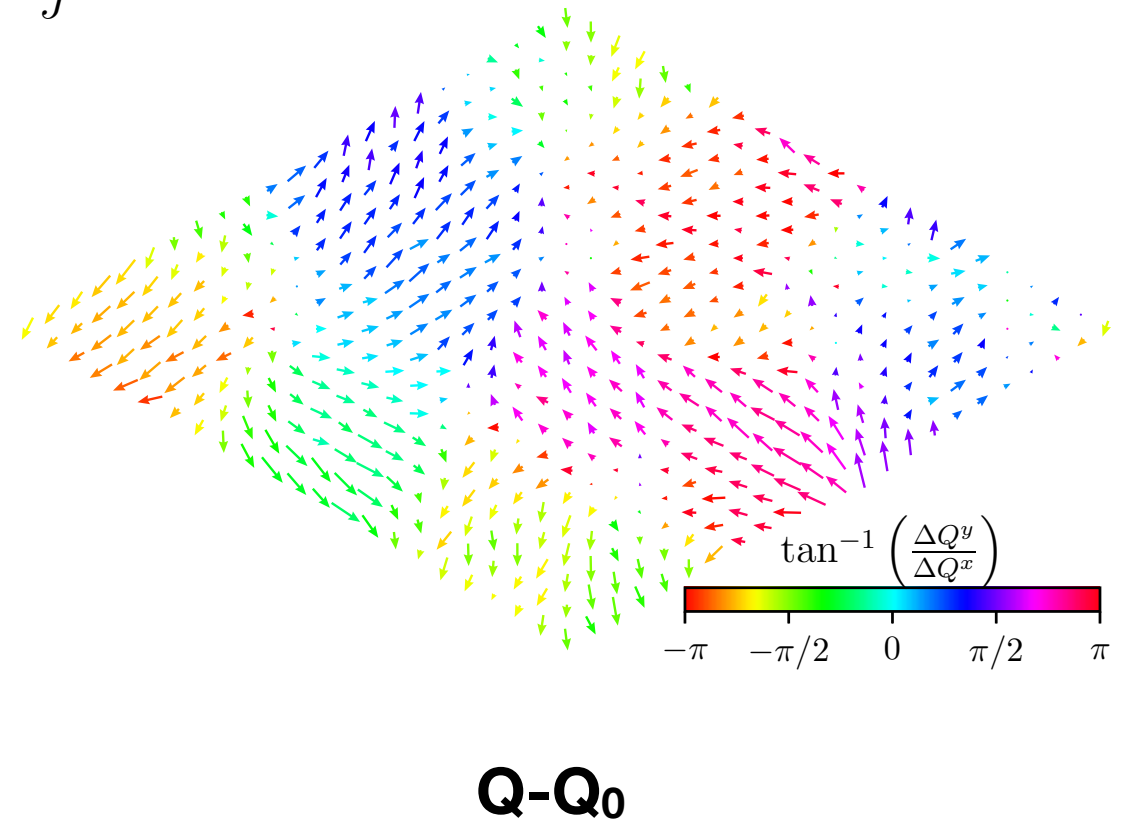
Numerical simulation

- Glassy state demarcated by domains of incommensurability. **Q-Q₀**

$$\frac{\delta J}{J}=0.1$$



$$\frac{\delta J}{J}=0.4$$



- Domain length decreases with increasing disorder.